

Survey Proposal: Reinforcement Learning on Graphs and its Applications

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1 Topic for Survey

Reinforcement Learning (RL) has traditionally excelled in domains with fixed, grid-like state spaces (such as images in Atari games) or low-dimensional vector spaces (such as robotic joint angles). However, many real-world problems are structurally irregular and relationally complex, naturally represented as graphs. These include transportation networks, power grids, molecular structures, and distributed sensor networks.

This survey examines "Graph Reinforcement Learning". It is a methodological framework that integrates Graph Neural Networks (GNNs) with RL agents. The core motivation for this integration is twofold. First, standard RL architectures (MLPs, CNNs) struggle to capture the relational inductive biases necessary to understand complex dependencies between entities. Second, real-world graphs often vary in size and topology; a policy trained on a 10-node network must ideally generalize to a 100-node network without retraining—a capability known as size-invariant generalization.

We review recent advancements in this field, categorizing approaches by their application domains: Combinatorial Optimization (e.g., vehicle routing, job shop scheduling), Network Control (e.g., power grids), and Embodied Agents (e.g., robot navigation). We specifically analyze how graph attention mechanisms and hierarchical structures are being used to solve the "curse of dimensionality" in large-scale state spaces.

2 Research Papers

1. **Reinforcement Learning on Graphs: A Survey (ArXiv 2022).**
<https://arxiv.org/abs/2204.06127>. Comprehensive taxonomy of Graph RL (GRL); categorizes methods by algorithmic structure and application domains like mechanics and chemistry.

2. **Graph Reinforcement Learning for Combinatorial Optimization: A Survey and Unifying Perspective (ArXiv 2024).**
<https://arxiv.org/abs/2404.06492>. Unifies "Graph RL" frameworks for decision-making; distinguishes between Structure Optimization (finding graphs) and Process Optimization (controlling dynamics).
3. **Graph Reinforcement Learning for Power Grids: A Comprehensive Survey (Energy & AI 2025).**
<https://arxiv.org/abs/2407.04522>. Systematic review of GNN-RL for transmission and distribution grids; highlights gaps in safety guarantees and real-world deployability.
4. **Graph Neural Networks for Job Shop Scheduling Problems: A Survey (ArXiv 2024).**
<https://arxiv.org/abs/2406.14096>. Survey of GNN-based schedulers; focuses on handling heterogeneity and scalability in NP-hard production environments.
5. **Factored Value Functions for Graph-Based Multi-Agent Reinforcement Learning (ArXiv 2026).**
<https://arxiv.org/abs/2601.11401>. Proposes "Diffusion Value Functions" (DVF) to decompose global rewards; improves credit assignment via graph diffusion in large-scale MARL.
6. **Dual-Attention Heterogeneous GNN for Multi-robot Collaborative Area Search (ArXiv 2026).**
<https://arxiv.org/abs/2601.03686>. Balances exploration vs. target coverage using dual-attention (relational + type-aware) on heterogeneous graphs.
7. **Hierarchical GNN-Based Multi-Agent Learning for Dynamic Queue-Jump Lane (ArXiv 2026).**
<https://arxiv.org/abs/2601.04177>. Hierarchical planner coordinating global strategy and local control for emergency corridors; achieves near-zero collision rates.
8. **Mean-Field Control on Sparse Graphs: From Local Limits to GNNs (ArXiv 2026).**
<https://arxiv.org/abs/2601.21477>. Establishes theoretical locality for Mean-Field Control on sparse graphs; proves validity of GNN-based actor-critics for scalable control.
9. **EGAM: Extended Graph Attention Model for Solving Routing Problems (ArXiv 2026).**
<https://arxiv.org/abs/2601.21281>. Generalizes Graph Attention Models (GAM) by embedding both edges and nodes; outperforms baselines on highly constrained routing tasks.

10. **Learning General Policies with Policy Gradient Methods (KR 2023 / ArXiv 2025).**
<https://arxiv.org/abs/2512.19366>. Combines GNNs with policy gradients to learn domain-independent planning policies; addresses the generalization-optimality tradeoff.
11. **Vehicle Routing Problems via Quantum Graph Attention Network Deep RL (SOICT 2025).**
<https://arxiv.org/abs/2511.15175>. Hybrid Quantum-Classical GNN (Q-GAT) for VRP; reduces trainable parameters by 50% while maintaining routing efficiency.
12. **Platform-Agnostic RL Framework for Safe Exploration with Graph Attention (ArXiv 2025).**
<https://arxiv.org/abs/2511.15358>. Integrates GNN-based waypoint selection with a safety filter; guarantees collision-free exploration in cluttered environments.
13. **Traffic-Aware Optimal Taxi Placement Using GNN-Based RL (ArXiv 2026).**
<https://arxiv.org/abs/2601.00607>. Encodes real-time traffic congestion into GNN embeddings; optimizes taxi hotspots to minimize passenger wait times.
14. **Socially Aware Navigation for Mobile Robots: A Survey on Deep RL (ArXiv 2025).**
<https://arxiv.org/abs/2512.00049>. Critical review of DRL for social navigation; identifies lack of standardized metrics and sim-to-real transfer as key bottlenecks.
15. **Graph Contextual Reinforcement Learning for Efficient Directed Controller Synthesis (ArXiv 2025).**
<https://arxiv.org/abs/2512.15295>. Uses GNNs to encode exploration history (GCRL); guides controller synthesis more efficiently than local-context RL methods.

3 Future Scope and Refinement

While this initial literature review surveys the broad spectrum of Graph Reinforcement Learning, ranging from static combinatorial optimization to dynamic multi-agent control, our final report will adopt a more targeted scope. Following this preliminary landscape analysis, we intend to narrow our focus to a specific sub-domain (likely Multi-Agent Path Finding or Distributed Network Control) to ensure a rigorous and deep comparative analysis of the final 15 selected papers.